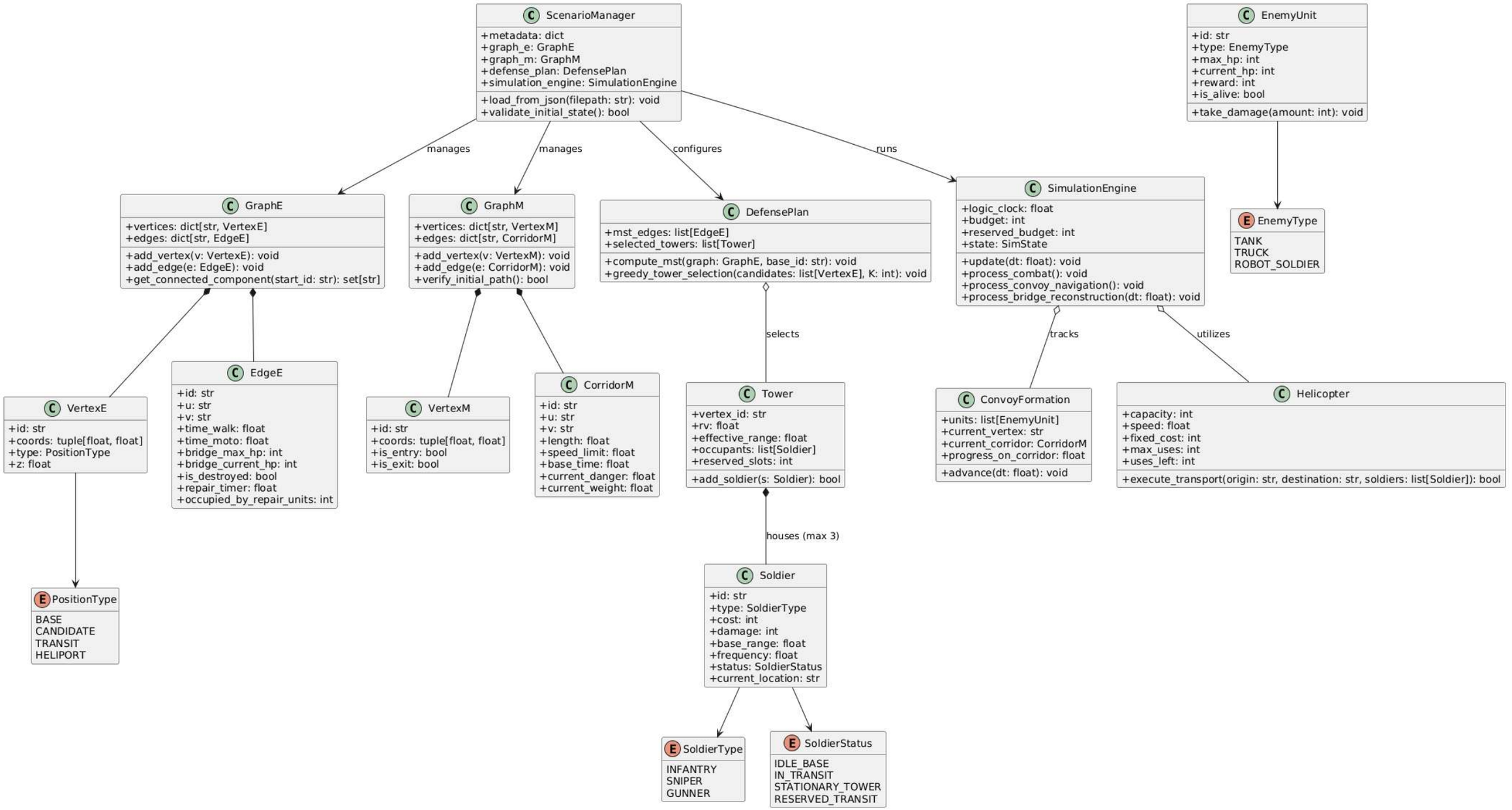

ScenarioManager
+metadata: dict
+graph_e: GraphE
+graph_m: GraphM
+defense_plan: DefensePlan
+simulation_engine: SimulationEngine
+load_from_json(filepath: str): void
+validate_initial_state(): bool
EnemyUnit
+id: str
+type: EnemyType
+max_hp: int
+current_hp: int
+reward: int
+is_alive: bool
+take_damage(amount: int): void
manages
manages
configures
runs
GraphE
+vertices: dict[str, VertexE]
+edges: dict[str, EdgeE]
+add_vertex(v: VertexE): void
+add_edge(e: EdgeE): void
+get_connected_component(start_id: str): set[str]
GraphM
+vertices: dict[str, VertexM]
+edges: dict[str, CorridorM]
+add_vertex(v: VertexM): void
+add_edge(e: CorridorM): void
+verify_initial_path(): bool
DefensePlan
+mst_edges: list[EdgeE]
+selected_towers: list[Tower]
+compute_mst(graph: GraphE, base_id: str): void
+greedy_tower_selection(candidates: list[VertexE], K: int): void
SimulationEngine
+logic_clock: float
+budget: int
+reserved_budget: int
+state: SimState
+update(dt: float): void
+process_combat(): void
+process_convoy_navigation(): void
+process_bridge_reconstruction(dt: float): void
EnemyType
TANK
TRUCK
ROBOT_SOLDIER
VertexE
+id: str
+coords: tuple[float, float]
+type: PositionType
+z: float
EdgeE
+id: str
+u: str
+v: str
+time_walk: float
+time_moto: float
+bridge_max_hp: int
+bridge_current_hp: int
+is_destroyed: bool
+repair_timer: float
+occupied_by_repair_units: int
VertexM
+id: str
+coords: tuple[float, float]
+is_entry: bool
+is_exit: bool
CorridorM
+id: str
+u: str
+v: str
+length: float
+speed_limit: float
+base_time: float
+current_danger: float
+current_weight: float
selects
Tower
+vertex_id: str
+rv: float
+effective_range: float
+occupants: list[Soldier]
+reserved_slots: int
+add_soldier(s: Soldier): bool
tracks
utilizes
ConvoyFormation
+units: list[EnemyUnit]
+current_vertex: str
+current_corridor: CorridorM
+progress_on_corridor: float
+advance(dt: float): void
Helicopter
+capacity: int
+speed: float
+fixed_cost: int
+max_uses: int
+uses_left: int
+execute_transport(origin: str, destination: str, soldiers: list[Soldier]): bool
PositionType
BASE
CANDIDATE
TRANSIT
HELIPORT
houses (max 3)
Soldier
+id: str
+type: SoldierType
+cost: int
+damage: int
+base_range: float
+frequency: float
+status: SoldierStatus
+current_location: str
SoldierType
INFANTRY
SNIPER
GUNNER
SoldierStatus
IDLE_BASE
IN_TRANSIT
STATIONARY_TOWER
RESERVED_TRANSIT